Commodore VIC-20 — Chip Pinouts

MAJOR SOCKETED ICS · CPU · VIC · VIA · RAM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for signal notation & the colour key. **6560**=NTSC / **6561**=PAL.

Legend — reading the pinouts (notation)

/NAME **Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME **Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: Vcc / Vdd = power · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

6502 CPU (40-DIP)

Signal	Pin	Pin	Signal
Vss	1	40	/RES
RDY	2	39	φ2 out
φ1 out	3	38	SO
/IRQ	4	37	φ0 in
NC	5	36	NC
/NMI	6	35	NC
SYNC	7	34	R/W
Vcc	8	33	D0
A0	9	32	D1
A1	10	31	D2
A2	11	30	D3
A3	12	29	D4
A4	13	28	D5
A5	14	27	D6
A6	15	26	D7
A7	16	25	A15
A8	17	24	A14
A9	18	23	A13
A10	19	22	A12
A11	20	21	Vss

VIC-20 CPU @ ~1 MHz (1.02 NTSC / 1.11 PAL). **Fails:** black/blank screen, dead machine.

VIC 6560/61 video (40-DIP)

Signal	Pin	Pin	Signal
NC	1	40	Vcc
CHROMA	2	39	φ1 in
SYNC/LUM	3	38	φ2 in
R/W	4	37	LP
VD11	5	36	φ out
VD10	6	35	φ1 out
VD9	7	34	A13
VD8	8	33	A12
D7	9	32	A11
D6	10	31	A10
D5	11	30	A9
D4	12	29	A8
D3	13	28	A7
D2	14	27	A6
D1	15	26	A5
D0	16	25	A4
POTX	17	24	A3
POTY	18	23	A2
AUDIO	19	22	A1
GND	20	21	A0

Video + 3 tone + noise sound + paddle A/D (POTX/Y) + light pen. **6560**=NTSC, **6561**=PAL @ $9000. **Fails:** no/garbled picture, no sound.

6522 VIA ×2 (40-DIP)

Signal	Pin	Pin	Signal
Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	RS0
PA2	4	37	RS1
PA3	5	36	RS2
PA4	6	35	RS3
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	φ2
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	R/W
Vcc	20	21	/IRQ

VIA1 $9110: /RESTORE→NMI, serial bus, cassette, user-port CB. **VIA2** $9120: keyboard scan, joystick, timers, IRQ. **Fails:** dead keyboard / joystick / serial / cassette.

2114 SRAM 1K×4 (18-DIP)

Signal	Pin	Pin	Signal
A6	1	18	Vcc
A5	2	17	A7
A4	3	16	A8
A3	4	15	A9
A0	5	14	D1
A1	6	13	D2
A2	7	12	D3
/CS	8	11	D4
GND	9	10	/WE

Main RAM (static). The unexpanded VIC-20 has 5KB built from these. **Fails:** garbled screen, won't boot, RAM errors.

ROMs (mask ROM)

Character	4K char generator
BASIC	8K — $C000
KERNAL	8K — $E000

Replace with 2364-compatible EPROM via adapter. **Fails:** garbage on boot, bad characters.

Test & Repair (tips)

Power	+5V logic only (no +12V rail)
Black screen	CPU / clock / KERNAL ROM
Garbage screen	2114 RAM or VIC
No picture/sound	VIC 6560/61
Dead keys/joystick	VIA2
No serial/cassette	VIA1

Socket suspect chips; swap order **VIC → CPU → RAM → VIAs**. The VIC-20 runs on +5V only.